

Day-10 Assignment

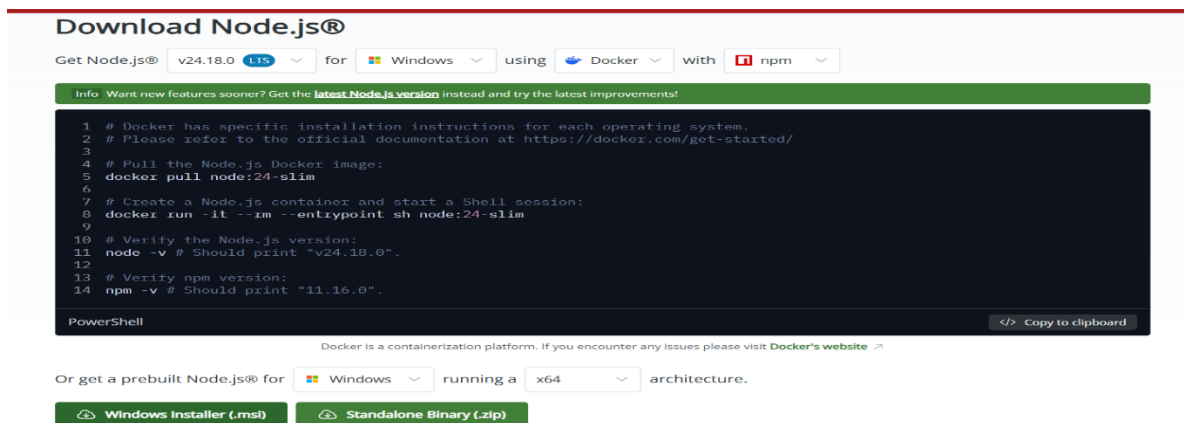
REST API Mini Project : Employee Management System

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Emp Id: 85010930

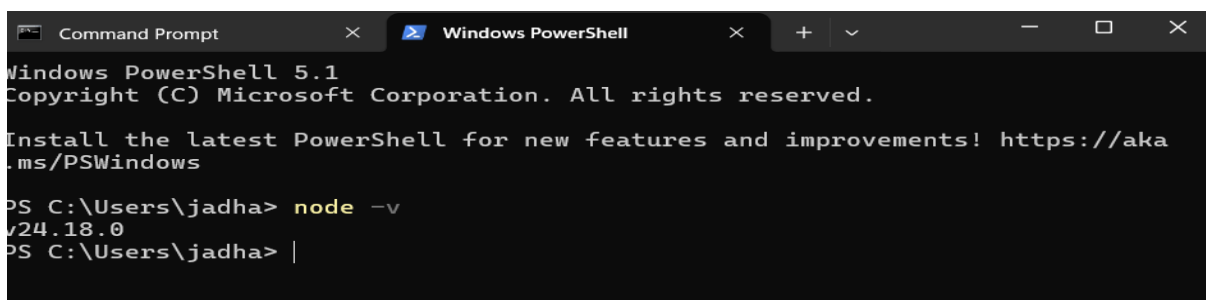
Step 1: Download and Install Node.js

Explanation: Node.js is a JavaScript runtime environment required to run JSON Server. It also installs npm (Node Package Manager), which is used to install JSON Server.



Step 2: Verify Node.js Installation

Explanation: After installing Node.js, verify the installation by checking the installed version using the command prompt or PowerShell



Step 3: Install JSON Server

Explanation: JSON Server is used to create a fake REST API from a JSON file. It helps developers test REST APIs without creating a backend application.

```
Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\EmployeeRestAPI>npm install -g json-server

changed 43 packages in 11s

15 packages are looking for funding
  run `npm fund` for details

C:\EmployeeRestAPI>|
```

Step 4: Verify JSON Server Installation

Explanation:

Verify that JSON Server is installed successfully by checking its version.

```
C:\EmployeeRestAPI>json-server --watch Employee.json --port 3000
--watch/-w can be omitted, JSON Server 1+ watches for file changes by default
JSON Server started on PORT :3000
Press CTRL-C to stop
Watching Employee.json...

(-_-)

Index:
http://localhost:3000/

Static files:
Serving ./public directory if it exists

Endpoints:
http://localhost:3000/employees
```

Step 5: Verify JSON Server in Browser

Explanation:

After starting JSON Server successfully, open the API endpoint in a web browser to verify that the server is running and returning employee data in JSON format.



```
[
  {
    "id": "101",
    "name": "Rahul Sharma",
    "department": "IT",
    "designation": "Software Engineer",
    "salary": 60000,
    "email": "rahul@gmail.com",
    "phone": "9876543210"
  },
  {
    "id": "102",
    "name": "Anjali",
    "department": "HR",
    "designation": "HR Executive",
    "salary": 45000,
    "email": "anjali@gmail.com",
    "phone": "9876543211"
  },
  {
    "id": "103",
    "name": "Sai",
    "department": "Finance",
    "designation": "Accountant",
    "salary": 50000,
    "email": "sai@gmail.com",
    "phone": "9876543212"
  }
]
```

Step 6: Download and Install Postman

Explanation:

Postman is an API testing tool used to send HTTP requests such as GET, POST, PUT, PATCH, and DELETE. It helps developers test REST APIs without writing frontend code.

Procedure:

1. Open your browser.
2. Visit <https://www.postman.com/downloads/>
3. Download the Windows Desktop Installer.
4. Install Postman by following the installation wizard.
5. Launch the Postman application.


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Download Postman

Download the desktop app for the most complete Postman experience, with native Git support and deeper integration into your development workflow.

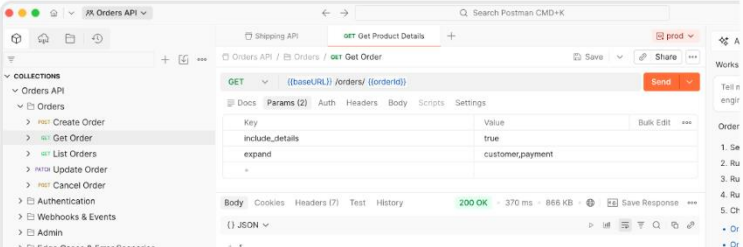
The Postman app

Download the app to get started with the Postman API Platform.

Check your inbox

We sent a download link for **Windows x64** to **jadhavavinash9390@gmail.com**. The link typically arrives in under a minute.

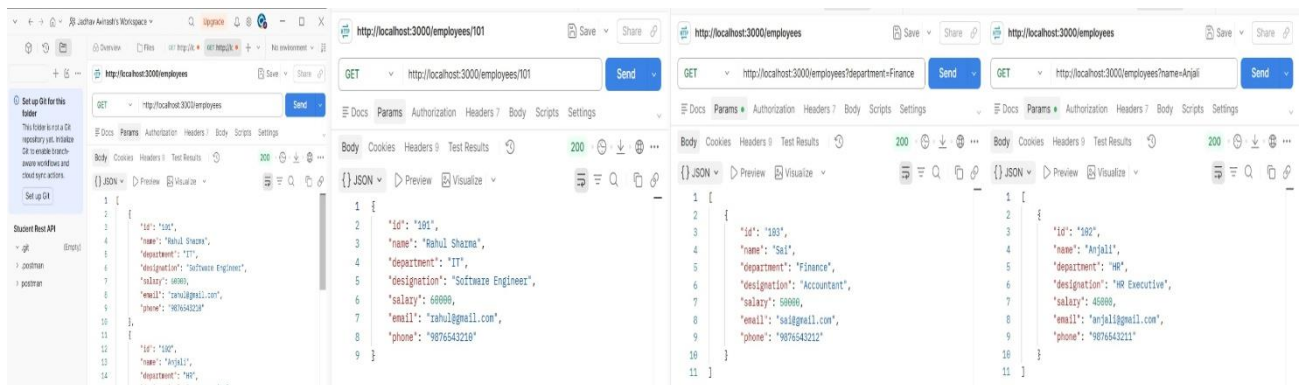
[Release Notes](#)



Step 7: Test GET REST API Endpoints Using Postman

Explanation:

The **GET** HTTP method is used to retrieve employee information from the JSON Server. Different GET endpoints were tested to retrieve all employees, a specific employee by ID, and employees based on search criteria such as department and name.



Step 8: POST – Add a New Employee

Explanation:

The **POST** method is used to create a new employee record by sending employee details in JSON format. After a successful request, the new employee is added to the JSON Server database.

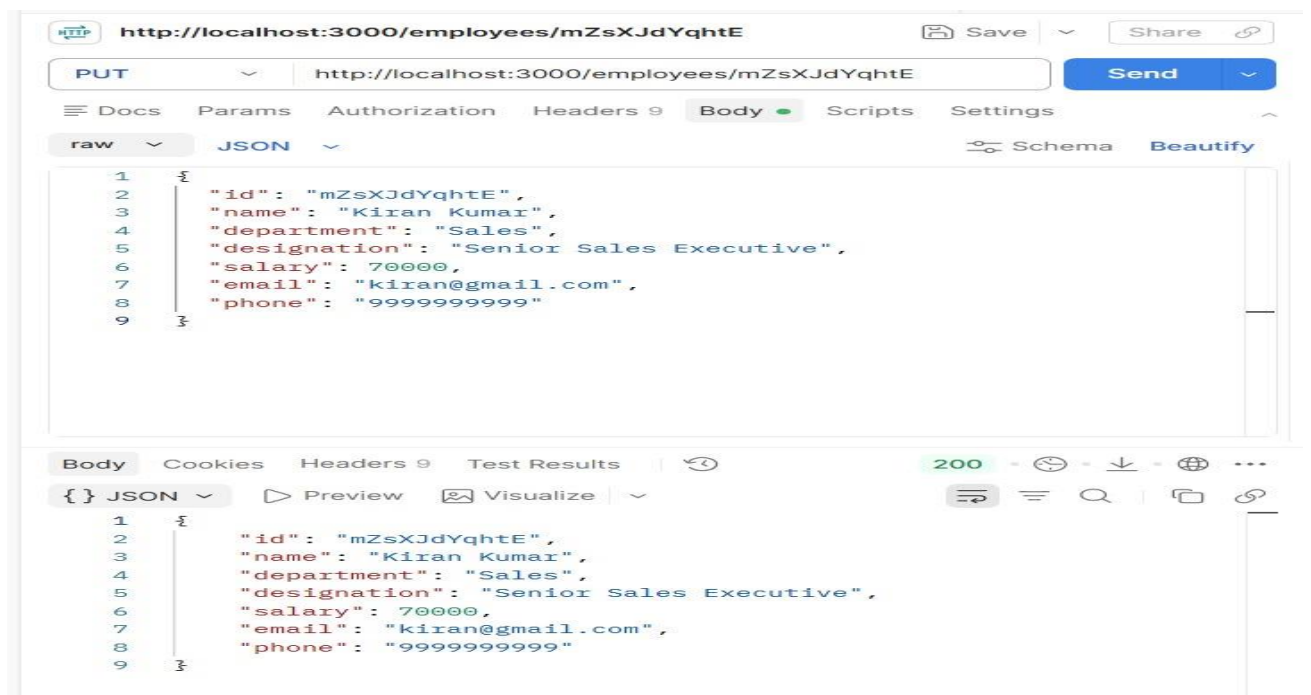
Method: POST



Step 9: PUT – Update Employee Details

Explanation: The PUT method is used to replace the complete employee record with updated information. It updates all fields of the selected employee.

Method: PUT

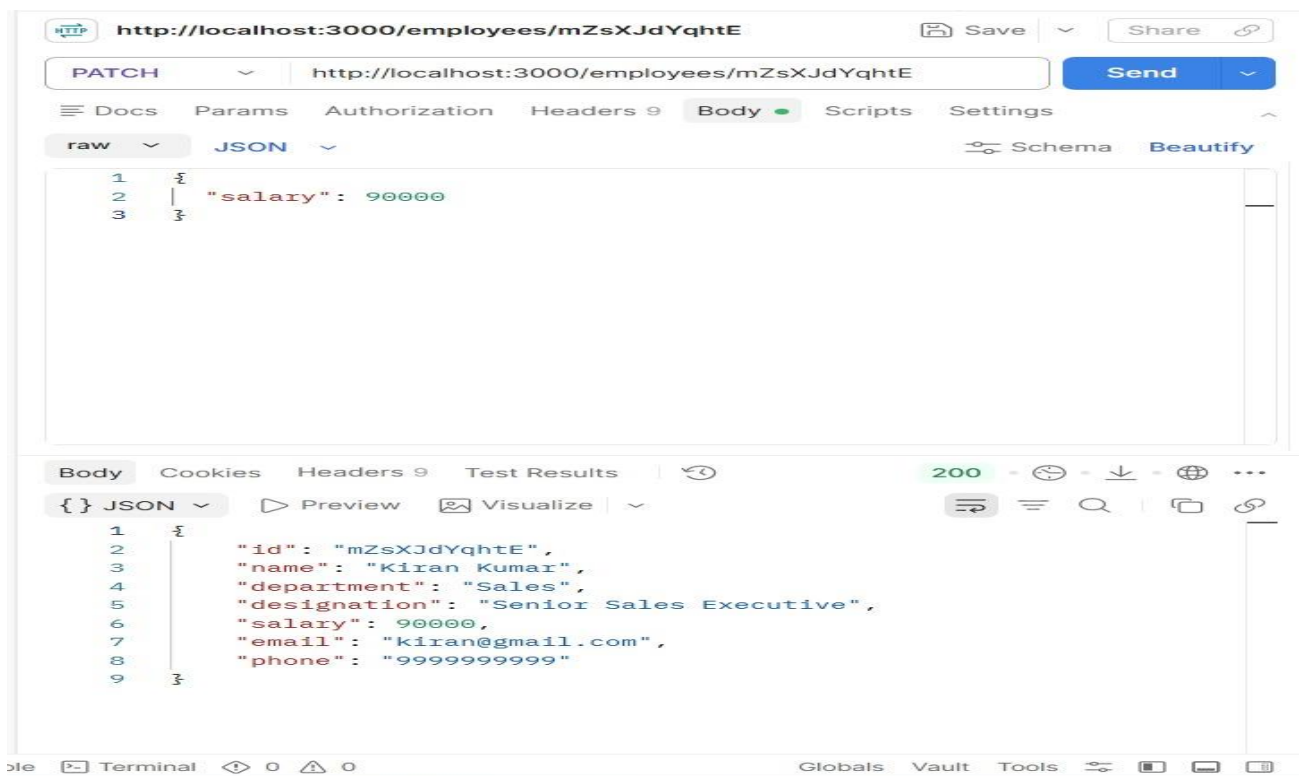


Step 10: PATCH – Update Employee Salary

Explanation:

The **PATCH** method is used to update only specific fields of an employee record. In this example, only the employee's salary is modified.

Method: PATCH



Result:

Only the salary field is updated, while the remaining employee details remain unchanged.

Step 11: DELETE – Remove Employee

Explanation:

The **DELETE** method is used to remove an employee record from the database using the employee's unique ID.

Method: DELETE



Result:

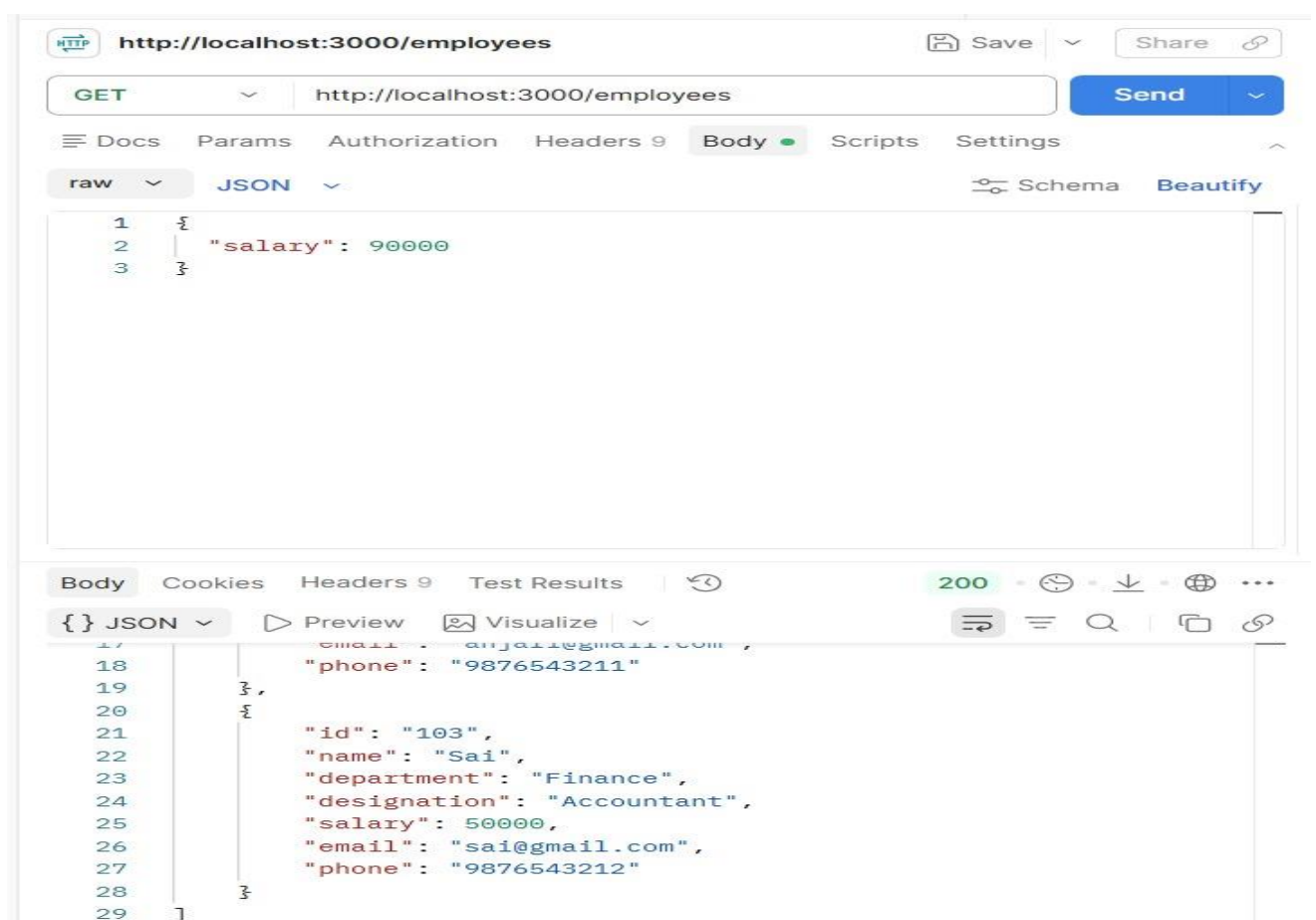
The employee record is deleted successfully, and the server returns **HTTP Status Code 200 (OK)**.

Step 12: Verify Employee Deletion

Explanation:

After deleting the employee, the **GET** method is used to retrieve all employee records and verify that the deleted employee no longer exists in the database.

Method: GET



Result: The employee Kiran Kumar is no longer displayed in the employee list, confirming that the DELETE operation was successful.

Conclusion:

I successfully configured JSON Server and created a REST API for the Employee Management System. I tested all CRUD operations using Postman, including Create, Read, Update, and Delete. This project improved my understanding of REST APIs, HTTP methods, JSON, and API testing, providing hands-on